DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
UNIVERSITY OF BARISHAL
Course Title: Data Structures
Course Code: CSE-1201
1st Year 2nd Semester
Session: 2021-222

Time: 3 Hours

Marks: 60

(Answer any FIVE questions)

1. a) Write down the algorithm to find the largest item from a list of data. 2

   b) Suppose, the department of CSE, BU keeps CGPA of final semester students in a linear array RESULT [1:100]. Now the chairperson of CSE wants to see an array with CGPA of ascending order. Select and write a sorting algorithm with O(nlogn) for all cases of the algorithm. 4

   c) Suppose we have an array 4
   DATA [] = {12, 10, 29, 24, 39, 51, 40, 56, 69}.
   Apply binary search algorithm to find the ITEM = 56 and ITEM = 52. How many times you have calculate MID value?

   d) State the function floor and ceil with proper example. 2

2. a) Define deque and priority queue. Write down the one-way representation of the following person according to their rank. 3

Person	Karim	Rahim	Sohan	Shakib	Mahmud
Rank	2	3	5	1(highest)	2

   b) State queues representation like array in memory. Show the insertion and deletion in case of circular queue. 3

   c) Suppose a STACK contains CSE, Physics, EEE (where EEE is the top). How many PUSH and POP operations are required for arranging the STACK like CSE, EEE, Physics (Physics in top)? Explain your answer with the respective figure if necessary. 4

   d) Write down the application of stack with real life examples. 2

3. a) Implement a simulation program for various operations on a linked list. The program should allow users to perform operations such as insertion, deletion, searching, and traversal on a linked list. Choose a specific type of linked list (e.g., singly linked list, doubly linked list) for this simulation. [10]
   Requirements:
   Insertion:
   i. Implement functions to insert nodes at the beginning, end, and at a specified position in the linked list.
   ii. Provide a visual representation of the linked list after each insertion operation.

   Deletion:
   i. Implement functions to delete nodes from the beginning, end, and at a specified position in the linked list.
   ii. Display the linked list after each deletion operation.

   Searching:
   i. Implement a search function that allows users to find a specific element in the linked list.
   ii. Display whether the element was found or not.

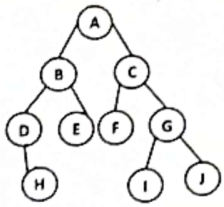
   Traversal:
   i. Implement a function to traverse and display the elements of the linked list.
   ii. Ensure a clear display of the linked list after each traversal operation.

   Random Operations
   i. Allow users to perform a combination of insertion, deletion, and searching operations in a randomized manner.
   ii. Demonstrate the linked list's state after each set of random operations.

   b) Define stack and Queue. Write down the push and pop operation for a Stack. [2]

AVAILABLE AT:
Onebyzero Edu - Organized Learning, Smooth Career
The Comprehensive Academic Study Platform for University Students in Bangladesh (www.onebyzeroedu.com)

4. a) Define complete binary tree. Traverse the given tree using In-order, Preorder and Post-order traversals.



[3]

b) Implement a program that allows users to perform basic operations on a Binary Search Tree (BST). The program should include functions for:

i. Insertion of nodes into the BST.
ii. Deletion of nodes from the BST.
iii. Searching for a specific value in the BST.
iv. In-order traversal of the BST.

c) Explain the concept of AVL Trees and their importance in maintaining balance. Discuss the rotations used in AVL Trees to restore balance during insertions and deletions. [2]

d) Compare the time complexity of operations in an AVL Tree with that in a regular BST. [5]

5. a) What do you mean by direct addressing table and hash table? [2]

b) Consider a hash table of size 7 with hash function $h(k) = k \mod 7$. Draw the table that results after inserting in the given order, the values, 19, 26, 13, 48, 17 for each of the scenarios below: [4]

i) When collisions are handled by separate chaining.
ii) When collisions are handled by linear probing.
iii) When collisions are handled by double hashing.
iv) Using a second hash function $h'(k) = 5 - (k \mod 5)$.

c) **What do you mean by disjoint set data structure?** Suppose that CONNECTED-COMPONENTS is run on the undirected graph G=(V, E), where V = { a, b, c, d, e, f, g, h, i, j, k} and the edges of E are processed in the order (d, i), (f, k), (g, i), (b, g), (a, h), (i, j), (d, k), (b, j), (d, f), (g, j), (a, e). **List the vertices in each connected component after each iteration of lines 3–5.** [4]

```
CONNECTED-COMPONENTS(G)
1  for each vertex v ∈ G.V
2      MAKE-SET(v)
3  for each edge (u, v) ∈ G.E
4      if FIND-SET(u) ≠ FIND-SET(v)
5          UNION(u, v)
```

d) Write down the union finding algorithm. [2]

6. a) Write down the number of hits does the following string matching algorithms encounter in the text ***T= abababababacba*** when looking for the pattern ***P= ababacc*** and also write down their procedure. [3]

i. Naive String Matching
ii. Finite State Automata

b) What kind of problem can be solved using dynamic programming algorithm. [4]

c) Define divide and conquer. Write down the pseudo code for finding the result of 7^{54} % 8 using divide and conquer. [3]

d) Represent the *tower of hanoi* problem using master theorem of recurrence. [2]

7. a) Consider the following adjacent list for directed graph. [5]
adj(y) = [x], adj(x) = [z], adj(z) = [y,w], adj(w) = [x], adj(s) = [z,w], adj(v) = [s,w], adj(t) = [u,v], adj(u) = [v].

i. Draw a graph from the above information
ii. Draw the adjacency and incidence matrix for the above graph. What are the advantages of adjacency matrix over adjacency list for graph traversal?
iii. Identify the tree edge, back edge, cross edge and forward edge for the graph.

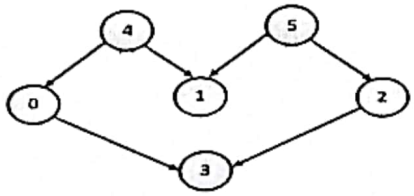
b) Write down the algorithm for the Breadth First Search with suitable illustration. [3]

AVAILABLE AT:
Onebyzero Edu - Organized Learning, Smooth Career
The Comprehensive Academic Study Platform for University Students in Bangladesh (www.onebyzeroedu.com)

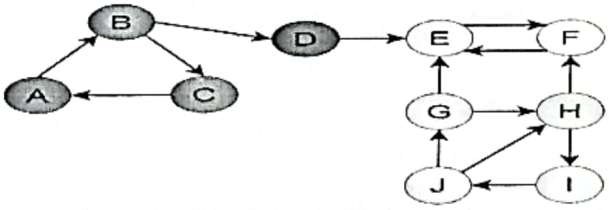
c) Topological sort algorithm for Directed Acyclic Graph (DAG) is a linear arrangement of vertices such that for every directed edge x-y, vertex x comes before y in the arrangement. Topological sorting is only applicable on graphs that are DAG. Now your task is to prove that from the above graph that, "There can exist more than one topological sorting for a given graph." Using [4]

i. DFS

ii. Kahn's Algorithm

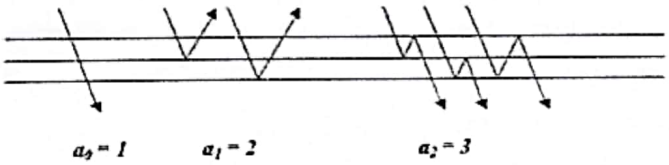


8. a) Find out the Strongly Connected Components (SCC) algorithms using DFS for the following figure. [4]



b) Consider two sequences: $X = (A, B, C, B, A, B)$ and $Y = (B, C, C, A, B, A)$. Find a maximum length common subsequence of X and Y. [**Hints:** Solve the problem using dynamic programming approach]. [2]

c) A well renowned UVA Problem: 10334, Suppose we put two panes of glass back-to-back. How many ways a_n are there for light rays to pass through or be reflected after changing direction n times? Following figure shows the situations when the value of n is 0, 1 and 2. Now your task is to write down the program for input 0, 1, 2: output will be 1, 2, and 3 respectively. [3]



d) Why do we need postfix and prefix notation over infix notation? Translate the infix notation $A*(B+D)/E-F*(G+H/K)$ to its equivalent postfix notation using stack. [3]

AVAILABLE AT:
Onebyzero Edu - Organized Learning, Smooth Career
The Comprehensive Academic Study Platform for University Students in Bangladesh (www.onebyzeroedu.com)